

値域が通常と異なる関数

$$\begin{cases} \arctan(x) : \mathbb{R} \mapsto [0, \pi) \\ \operatorname{atan2}(y, x) : \mathbb{R}^2 \setminus \{(0, 0)\} \mapsto [0, 2\pi) \end{cases}$$

1 2次元直交座標

$$\begin{cases} \mathbf{e}_x = (1, 0) \\ \mathbf{e}_y = (0, 1) \end{cases}$$

$\mathbf{e}_x, \mathbf{e}_y$ は t に対して不変 (定数のように扱われる)

$$\begin{aligned} \mathbf{r}(x(t), y(t)) &= x(t)\mathbf{e}_x + y(t)\mathbf{e}_y \\ \frac{d\mathbf{r}(x(t), y(t))}{dt} &= \frac{dx(t)}{dt}\mathbf{e}_x + \frac{dy(t)}{dt}\mathbf{e}_y \\ \frac{d^2\mathbf{r}(x(t), y(t))}{dt^2} &= \frac{d^2x(t)}{dt^2}\mathbf{e}_x + \frac{d^2y(t)}{dt^2}\mathbf{e}_y \end{aligned}$$

2 基底ベクトルの変換

直交座標からの変数変換

$$(x_1(t), \dots, x_n(t)) \rightarrow (x_1((y_1(t), \dots, y_n(t))), \dots, x_n((y_1(t), \dots, y_n(t))))$$

により

$$\mathbf{r}(y_1(t), \dots, y_n(t)) = \sum_{k=1}^n x_k(y_1(t), \dots, y_n(t))\mathbf{e}_{x_k}$$

として表現可能な場合、各変数方向の基底ベクトルは

$$\begin{aligned} \mathbf{e}_{y_k}(y_1(t), \dots, y_n(t)) &= \frac{\partial_{y_k} \mathbf{r}(y_1(t), \dots, y_n(t))}{\left| \partial_{y_k} \mathbf{r}(y_1(t), \dots, y_n(t)) \right|} \\ &= \frac{\partial_{y_k} \mathbf{r}(y_1(t), \dots, y_n(t))}{\sqrt{\sum_{i=1}^n \partial_{y_k} x_i^2(y_1(t), \dots, y_n(t))}} \end{aligned}$$

で表される。

3 2次元極座標

基本の変換

$$\begin{cases} x(r(t), \theta(t)) = r(t) \cos \theta(t) \\ y(r(t), \theta(t)) = r(t) \sin \theta(t) \end{cases}$$
$$\begin{cases} r(x(t), y(t)) = \sqrt{x^2(t) + y^2(t)} \\ \theta(x(t), y(t)) = \operatorname{atan2}(y(t), x(t)) \end{cases}$$

2次元極座標では

$$\mathbf{r}(r(t), \theta(t)) = x(r(t), \theta(t))\mathbf{e}_x + y(r(t), \theta(t))\mathbf{e}_y = r(t) \cos \theta(t)\mathbf{e}_x + r(t) \sin \theta(t)\mathbf{e}_y$$

の関係を用いると、

$$\mathbf{e}_r(r(t), \theta(t)) = \cos \theta(t)\mathbf{e}_x + \sin \theta(t)\mathbf{e}_y$$

$$\mathbf{e}_\theta(r(t), \theta(t)) = -\sin \theta(t)\mathbf{e}_x + \cos \theta(t)\mathbf{e}_y$$

として表される。次に、各基底ベクトルの時間微分を求める。多変数関数の微分

$$\frac{df(x_1, \dots, x_n)}{dt} = \sum_{i=1}^n \frac{\partial f(x_1, \dots, x_n)}{\partial x_i} \frac{dx_i}{dt}$$

を用いると、 $r(t)$ での偏微分は0になり、

$$\begin{aligned} \frac{d\mathbf{e}_r(r(t), \theta(t))}{dt} &= -\sin \theta(t) \frac{d\theta(t)}{dt} \mathbf{e}_x + \cos \theta(t) \frac{d\theta(t)}{dt} \mathbf{e}_y \\ &= \frac{d\theta(t)}{dt} \mathbf{e}_\theta(r(t), \theta(t)) \\ \frac{d\mathbf{e}_\theta(r(t), \theta(t))}{dt} &= -\cos \theta(t) \frac{d\theta(t)}{dt} \mathbf{e}_x - \sin \theta(t) \frac{d\theta(t)}{dt} \mathbf{e}_y \\ &= -\frac{d\theta(t)}{dt} \mathbf{e}_r(r(t), \theta(t)) \end{aligned}$$

が得られる。この結果を用いて、位置ベクトルの微分を求めることができる。

(積の微分を用いる)

$$\begin{aligned} \mathbf{r}(r(t), \theta(t)) &= r(t)\mathbf{e}_r(r(t), \theta(t)) \\ \frac{d\mathbf{r}(r(t), \theta(t))}{dt} &= \frac{dr(t)}{dt} \mathbf{e}_r(r(t), \theta(t)) + r(t) \frac{d\mathbf{e}_r(r(t), \theta(t))}{dt} \\ &= \frac{dr(t)}{dt} \mathbf{e}_r(r(t), \theta(t)) + r(t) \frac{d\theta(t)}{dt} \mathbf{e}_\theta(r(t), \theta(t)) \\ \frac{d^2\mathbf{r}(r(t), \theta(t))}{dt^2} &= \frac{d}{dt} \left(\frac{dr(t)}{dt} \mathbf{e}_r(r(t), \theta(t)) + r(t) \frac{d\theta(t)}{dt} \mathbf{e}_\theta(r(t), \theta(t)) \right) \\ &= \frac{d^2r(t)}{dt^2} \mathbf{e}_r(r(t), \theta(t)) + \frac{dr(t)}{dt} \frac{d\mathbf{e}_r(r(t), \theta(t))}{dt} \\ &\quad + \frac{dr(t)}{dt} \frac{d\theta(t)}{dt} \mathbf{e}_\theta(r(t), \theta(t)) + r(t) \frac{d^2\theta(t)}{dt^2} \mathbf{e}_\theta(r(t), \theta(t)) + r(t) \frac{d\theta(t)}{dt} \frac{d\mathbf{e}_\theta(r(t), \theta(t))}{dt} \\ &= \frac{d^2r(t)}{dt^2} \mathbf{e}_r(r(t), \theta(t)) + \frac{dr(t)}{dt} \frac{d\theta(t)}{dt} \mathbf{e}_\theta(r(t), \theta(t)) \\ &\quad + \frac{dr(t)}{dt} \frac{d\theta(t)}{dt} \mathbf{e}_\theta(r(t), \theta(t)) + r(t) \frac{d^2\theta(t)}{dt^2} \mathbf{e}_\theta(r(t), \theta(t)) - r(t) \left(\frac{d\theta(t)}{dt} \right)^2 \mathbf{e}_r(r(t), \theta(t)) \\ &= \left(\frac{d^2r(t)}{dt^2} - r(t) \left(\frac{d\theta(t)}{dt} \right)^2 \right) \mathbf{e}_r(r(t), \theta(t)) \\ &\quad + \left(2 \frac{dr(t)}{dt} \frac{d\theta(t)}{dt} + r(t) \frac{d^2\theta(t)}{dt^2} \right) \mathbf{e}_\theta(r(t), \theta(t)) \end{aligned}$$

4 3次元直交座標

$$\begin{cases} \mathbf{e}_x = (1, 0, 0) \\ \mathbf{e}_y = (0, 1, 0) \\ \mathbf{e}_z = (0, 0, 1) \end{cases}$$

$\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z$ は t に対して不変 (定数のように扱われる)

$$\begin{aligned} \mathbf{r}(x(t), y(t), z(t)) &= x(t)\mathbf{e}_x + y(t)\mathbf{e}_y + z(t)\mathbf{e}_z \\ \frac{d\mathbf{r}(x(t), y(t), z(t))}{dt} &= \frac{dx(t)}{dt}\mathbf{e}_x + \frac{dy(t)}{dt}\mathbf{e}_y + \frac{dz(t)}{dt}\mathbf{e}_z \\ \frac{d^2\mathbf{r}(x(t), y(t), z(t))}{dt^2} &= \frac{d^2x(t)}{dt^2}\mathbf{e}_x + \frac{d^2y(t)}{dt^2}\mathbf{e}_y + \frac{d^2z(t)}{dt^2}\mathbf{e}_z \end{aligned}$$

5 3次元極座標 (右手系)

基本の変換

$$\begin{cases} x(r(t), \theta(t), \varphi(t)) = r(t) \sin \theta(t) \cos \varphi(t) \\ y(r(t), \theta(t), \varphi(t)) = r(t) \sin \theta(t) \sin \varphi(t) \\ z(r(t), \theta(t), \varphi(t)) = r(t) \cos \theta(t) \end{cases}$$
$$\begin{cases} r(x(t), y(t), z(t)) = \sqrt{x^2(t) + y^2(t) + z^2(t)} \\ \theta(x(t), y(t), z(t)) = \arctan \frac{\sqrt{x^2(t) + y^2(t)}}{z(t)} \\ \varphi(x(t), y(t), z(t)) = \text{atan2}(y(t), x(t)) \end{cases}$$

やることは基本一緒なので、結果だけ載せる。

$$\mathbf{r}(r(t), \theta(t), \varphi(t)) = r(t) \sin \theta(t) \cos \varphi(t) \mathbf{e}_x + r(t) \sin \theta(t) \sin \varphi(t) \mathbf{e}_y + r(t) \cos \theta(t) \mathbf{e}_z$$

$$\mathbf{e}_r(r(t), \theta(t), \varphi(t)) = \sin \theta(t) \cos \varphi(t) \mathbf{e}_x + \sin \theta(t) \sin \varphi(t) \mathbf{e}_y + \cos \theta(t) \mathbf{e}_z$$

$$\mathbf{e}_\theta(r(t), \theta(t), \varphi(t)) = \cos \theta(t) \cos \varphi(t) \mathbf{e}_x + \cos \theta(t) \sin \varphi(t) \mathbf{e}_y - \sin \theta(t) \mathbf{e}_z$$

$$\mathbf{e}_\varphi(r(t), \theta(t), \varphi(t)) = -\sin \varphi(t) \mathbf{e}_x + \cos \varphi(t) \mathbf{e}_y$$

$$\frac{d\mathbf{e}_r(r(t), \theta(t), \varphi(t))}{dt} = \frac{d\theta(t)}{dt} \mathbf{e}_\theta(r(t), \theta(t), \varphi(t)) + \sin \theta(t) \frac{d\varphi(t)}{dt} \mathbf{e}_\varphi(r(t), \theta(t), \varphi(t))$$

$$\frac{d\mathbf{e}_\theta(r(t), \theta(t), \varphi(t))}{dt} = -\frac{d\theta(t)}{dt} \mathbf{e}_r(r(t), \theta(t), \varphi(t)) + \cos \theta(t) \frac{d\varphi(t)}{dt} \mathbf{e}_\varphi(r(t), \theta(t), \varphi(t))$$

$$\frac{d\mathbf{e}_\varphi(r(t), \theta(t), \varphi(t))}{dt} = -\sin \theta(t) \frac{d\varphi(t)}{dt} \mathbf{e}_r(r(t), \theta(t), \varphi(t)) - \cos \theta(t) \frac{d\varphi(t)}{dt} \mathbf{e}_\theta(r(t), \theta(t), \varphi(t))$$

$$\mathbf{r}(r(t), \theta(t), \varphi(t)) = r(t) \mathbf{e}_r(r(t), \theta(t), \varphi(t))$$

$$\begin{aligned} \frac{d\mathbf{r}(r(t), \theta(t), \varphi(t))}{dt} &= \frac{dr(t)}{dt} \mathbf{e}_r(r(t), \theta(t), \varphi(t)) + r(t) \frac{d\theta(t)}{dt} \mathbf{e}_\theta(r(t), \theta(t), \varphi(t)) + r(t) \sin \theta(t) \frac{d\varphi(t)}{dt} \mathbf{e}_\varphi(r(t), \theta(t), \varphi(t)) \\ \frac{d^2\mathbf{r}(r(t), \theta(t), \varphi(t))}{dt^2} &= \left(\frac{d^2r(t)}{dt^2} - r(t) \left(\frac{d\theta(t)}{dt} \right)^2 - r(t) \sin^2 \theta(t) \left(\frac{d\varphi(t)}{dt} \right)^2 \right) \mathbf{e}_r(r(t), \theta(t), \varphi(t)) \\ &\quad + \left(2 \frac{dr(t)}{dt} \frac{d\theta(t)}{dt} + r(t) \frac{d^2\theta(t)}{dt^2} - r(t) \sin \theta(t) \cos \theta(t) \left(\frac{d\varphi(t)}{dt} \right)^2 \right) \mathbf{e}_\theta(r(t), \theta(t), \varphi(t)) \\ &\quad + \left(2 \frac{dr(t)}{dt} \sin \theta(t) \frac{d\varphi(t)}{dt} + 2r(t) \cos \theta(t) \frac{d\theta(t)}{dt} \frac{d\varphi(t)}{dt} + r(t) \sin \theta(t) \frac{d^2\varphi(t)}{dt^2} \right) \mathbf{e}_\varphi(r(t), \theta(t), \varphi(t)) \end{aligned}$$

6 円筒座標 (右手系)

基本の変換

$$\begin{cases} x(\rho(t), \varphi(t), z(t)) = \rho(t) \cos \varphi(t) \\ y(\rho(t), \varphi(t), z(t)) = \rho(t) \sin \varphi(t) \\ z(\rho(t), \varphi(t), z(t)) = z(t) \end{cases}$$

$$\begin{cases} \rho(x(t), y(t), z(t)) = \sqrt{x^2(t) + y^2(t)} \\ \varphi(x(t), y(t), z(t)) = \text{atan2}(y(t), x(t)) \\ z(x(t), y(t), z(t)) = z(t) \end{cases}$$

円筒座標は 2 次元極座標と直交座標の和による合成である。

$$\mathbf{r}(\rho(t), \varphi(t), z(t)) = \rho(t) \cos \varphi(t) \mathbf{e}_x + \rho(t) \sin \varphi(t) \mathbf{e}_y + z(t) \mathbf{e}_z$$

$$\mathbf{e}_\rho(\rho(t), \varphi(t), z(t)) = \cos \varphi(t) \mathbf{e}_x + \sin \varphi(t) \mathbf{e}_y$$

$$\mathbf{e}_\varphi(\rho(t), \varphi(t), z(t)) = -\sin \varphi(t) \mathbf{e}_x + \cos \varphi(t) \mathbf{e}_y$$

$$\mathbf{e}_z(\rho(t), \varphi(t), z(t)) = \mathbf{e}_z$$

$$\frac{d\mathbf{e}_\rho(\rho(t), \varphi(t), z(t))}{dt} = \frac{d\varphi(t)}{dt} \mathbf{e}_\varphi(r(t), \theta(t), \varphi(t))$$

$$\frac{d\mathbf{e}_\varphi(\rho(t), \varphi(t), z(t))}{dt} = -\frac{d\varphi(t)}{dt} \mathbf{e}_\rho(r(t), \theta(t), \varphi(t))$$

$$\frac{d\mathbf{e}_z(\rho(t), \varphi(t), z(t))}{dt} = 0$$

$$\mathbf{r}(\rho(t), \varphi(t), z(t)) = \rho(t) \mathbf{e}_\rho(r(t), \theta(t), \varphi(t)) + z(t) \mathbf{e}_z$$

$$\frac{d\mathbf{r}(\rho(t), \varphi(t), z(t))}{dt} = \frac{d\rho(t)}{dt} \mathbf{e}_\rho(\rho(t), \varphi(t), z(t)) + \rho(t) \frac{d\varphi(t)}{dt} \mathbf{e}_\varphi(\rho(t), \varphi(t), z(t)) + \frac{dz(t)}{dt} \mathbf{e}_z$$

$$\frac{d^2\mathbf{r}(\rho(t), \varphi(t), z(t))}{dt^2} = \left(\frac{d^2\rho(t)}{dt^2} - \rho(t) \left(\frac{d\varphi(t)}{dt} \right)^2 \right) \mathbf{e}_\rho(\rho(t), \varphi(t), z(t))$$

$$+ \left(2 \frac{d\rho(t)}{dt} \frac{d\varphi(t)}{dt} + \rho(t) \frac{d^2\varphi(t)}{dt^2} \right) \mathbf{e}_\varphi(\rho(t), \varphi(t), z(t))$$

$$+ \frac{d^2z(t)}{dt^2} \mathbf{e}_z$$